ALAGAPPA UNIVERSITY JOURNAL OF BIOLOGICAL SCIENCES (AUJBS)

Vol 1 No 1 March 2017 ISSN:

Reach at

VI Floor Science Block
Alagappa University
Karaikudi 630 003 Tamil Nadu India
alubiojournal@gmail.com

Contents

S. No	Title	Page No.
1	Synthesis of silver nanoparticles of some edible basidiomycetes mushroom fungi using response surface methodology and its potential biological application **R Madhanraj, M Eyini and P Balaji**	01
2	Impact of CO_2 on growth, pigments yield and biochemical composition of marine microalga *Dunaliella salina* **A Shenbaga Devi, P Santhanam, S Jeyanthi, B Balaji Prasath and S Dinesh Kumar**	13
3	Fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus* **K Chinnadurai, M Eyini and P Balaji**	23
4	HPLC and biochemical techniques for secondary metabolites in *Garcinia indica* Choisy (Kokum) from transitional zones of Karnataka **Lingappa Sivakumar and Thirugnanasambandam Somasundaram**	35
5	Primary productivity of river chaliyar of Calicut district, Kerala, India **B Dhanalakshmi and P Priyatharsini**	48
6	Anti-bacterial activity, anti-inflammatory and anti- arthritic studies on mangroves by using in vitro model systems **M Babu Selvam and S Abideen**	54
7	Parasitic isopods of the family Cymothoidae from Indian fishes **S Ravichandran and G Ramesh Kumar**	65
8	Isolation and identification of pathogenic bacteria and its antibacterial susceptibility analysis in edible fish *Catla catla* **Mayavan Karthika, Shameem Shabana, Shamoon Muhasin and Venkatachalam Ramasubramanian**	72

9	Biogenic synthesis of silver nanoparticles from *Cardiospermum halicacabum* decorated with Graphene oxide for enhancing antibacterial ability **Gurusamy Sivaprakash, Gujuluva Hari Dinesh, Kulanthaisamy Mohan Rasu, Manoharan Dhivya and Alagarsamy Arun**	80
10	Studies on biosynthesis of xanthan gum using *Xanthomonas* sp., isolated from infected cotton leaves **V Ananthi and A Arun**	88
11	Characterization and determination of antibacterial activity of bacteriocin producing Lactic acid bacteria isolated from curd sample **V Ananthi and A Arun**	95
12	Antibacterial and immunostimulant influence of herbal extracts in grouper *Epinephelus tauvina* experimental culture against *Vibrio harveyi* Infection **T Citarasu, M Michael Babu and SMJ Punitha**	103
13	Assessment of bacteriological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water: potential threat of transmission of enteric pathogens and implications for public health **K Ramamoorthy and Clara G Sargunar**	117
14	Synthesis of chitin form shrimp dispel and its antibacterial activity **P Raja Rajeswari, R Shyamala Gowri, P Meenambigai and K Rajeswari**	132
15	Assessment of antibacterial activity of different solvent extracts of medicinal plant: *Aegle marmelos* **R Shyamala Gowri, R Vijayaraghavan, P Meenambigai and P Raja Rajeswari**	138
16	Effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of fresh water fish **S Chinniah, T Sangeetha and Subeena Begum**	145
17	A study on biologically synthesize silver nanoparticles using red seaweed *Gracilaria gracilis* **V Veeramanikandan, PT Usha and P Balaji**	154

Effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of fresh water fish

S Chinniah, T Sangeetha and **Subeena Begum***
Assistant Professor in Zoology, Department of Animal Health and Management,
Alagappa University, Karaikudi, Tamil Nadu, India

Received: 02.02.2017 / Accepted: 05.03.2017
Published online: 25.03.2017

Abstract *Tridax procumbens* is a wound healing medicinal plant. They are good immunostimulants. The present study is aimed to determine the effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of the fresh water fish *C. striatus.* Aqueous methanolic extract of *Tridax procumbens* is mixed thoroughly with the balanced diet at concentrations of 0.0%, 0.01%, 0.1%,1.0%. The prepared diets were fed to healthy *C. striatus* for 1- 4 weeks. The blood was drawn for every 7 days interval. Non-specific immune response such as Total WBC count and Phagocytic activity were analysed. Total WBC count and Phagocytic activity increased gradually in plant extract treated fish groups. There was a significant increase in the 3rd week of the experiment than in 2nd and 1st week. The same level was observed in the 4th week. The total WBC count and Phagocytic activity were increased in the plant extract treated fish than in control. The present study proved that the concentration of 1.0% aqueous methanolic extract of *Tridax procumbens* enhanced nonspecific immune responses.

Key Words *Tridax procumbens, Channa striatus,* Aqueous methanolic extract, Total WBC, Phagocytic activity.

Introduction

Aquaculture is the production of aquatic animals under the controlled condition Stickney *et, al.,(2000).*provides over 3% of the animal protein for the human consumption (WHO,2003, FAO 2010).Aquaculture is a major economic activity and an important source of foreign export for several development in our countries. It is estimated that freshwater fishes make up more than 6% of the world's annual animal protein supplies for humans (FAO, 2007). Aquaculture is contributing fish protein with minerals such as zinc, magnesium, sodium to the consumers Ravenhalt, 1982; and Barlas, (1986). Immunostimulants are substances which stimulate the specific and nonspecific defense systems of fish. It helps to enhance resistance to pathogens during stressful periods (Masoud Haghighi., and Mostafa Sharif Rohani (2013). Due to intensive culture practices for the increased production, disease management continues to pose a serious threat to aquaculture industry. Some plants are rich sources of compounds like volatile oils, saponins, phenolic compounds, tannins, alkaloids, polypeptides and polysaccharides. These natural plant products have various activities like antistress, appetizer, antimicrobials and immunostimulants (Citarasu *et al.,* 2003). The use of herbal immunostimulants through oral administration by the incorporation of it into aqua-feed is considered as a modern and promising alternative to antibiotics and vaccines.

*drsubeena2014@gmail.com

This study is being concentrated by many researchers. As the herbal immunostimulants are natural product, it does not produce any adverse effect to the aquatic environment and the organisms. Current trends in the use of medicinal plant extracts as an alternative to the drugs, chemicals and antibiotics in controlling fish diseases is growing because plant extracts, in contrast to vaccines, enhance the innate (or nonspecific) immune response (Sakai, 1999). Many herbal plants found to inhibit the bacterial pathogens and activate the immunity (Chansue *et al., 2000*; and Dugenci *et al*., 2003) at a low concentration and hence its use was very cost effective (Lipton, 2009).

The application of herbal plant extracts as a potential therapeutic measure for modulating the immune response in fish was found out; *Azadirachta indica* on *Channa striatus* (Udaya kumar and Venkatachalam., 2014); Achyranthus aspera on *Labeo rohita* (Rao *et al*., 2006); Withania somnifera on *Labeo rohita* (Sharma et al., 2010); Toona sinensis on *O. mossambicus* (Wu *et al*., 2010); neem formulation on *Cyprinus carpio* (Balasubramanian, 2006); *Cynodon dactylon* on *Catla catla* (Kaleeswaran *et al*.,2010); plant extract supplemented diet on *Cyprinus carpio* (Mohamed and Abasali., 2010).

The immunostimulatory effect of medicinal plants in fish were documented; *S. trilobatum* and *O. sanctum* in *M. keletius*., (Subeenabegum and Navaraj., 2016) *Ocimum sanctum* in *Cyprinus carpio* (Pavaraj *et al*., 2011); *Solanum trilobatum* (Divyagnaneswari *et al*., 2007); *Tinospora cordifolia* leaves (Sudhakaran *et al*., 2006; and Alexander *et al*., 2010); *Azadirachta indica* leaves on *C. carpio* (Wafaa *et al*., 2007). The measurement of total WBC count is the first signal to understand immune responses in fish administered with plant extracts. The total WBC counts increase by the plant extracts; *Allium sativum* on rainbow trout *Oncorhynchus mykiss* (Nya and Austin, 2009) were identified. Similar observations in other plant extracts; Fin – immune TM (a preparation of Cordyceps mushroom derived β glucan vitamins and essential minerals) in *Onchorhynchus mykiss* (Barker and Holliday, 2009); lipopolysaccharide in *Cyprinus carpio* (Selvaraj *et al*., 2006).

Phagocytosis has been recognized as an important activity in the host's defense against invading microorganisms (MacArthur *et al*, 1985; and Olivier *et al*., 1986). Fish were showing enhanced phagocytic activity due to various immunostimulants such as ginger (Dugenci *et al*., 2003) in rainbow trout; *Withania somnifera root* in *Labeo rohita* fingerlings (Sharma *et al*., 2010); *Ocimum sanctum* in *Cyprinus carpio* (Pavaraj *et al*., 2011); *Solanum trilobatum* in *Cyprinus carpio* (Durga devi and Balasubramanian, 2009).

Hitherto, the understanding about the immunostimulatory effect aqueous methanolic extract of *Tridax procumbens* on non-specific immune responses via., total WBC counts, phagocytic activity, in the fresh water fish *C. striatus* is very limited and hence this study. The main objective of this study was to determine the immunostimulatory effect of the phytocomponents present in the plant extract Tridax procumbens on the non-specific immune response in fish *C. striatus*.

Methods and materials

Fish: collection and maintenance

Clinically active fish purchased from local ponds from pudhuwayal near Karaikudi, Sivagangai District, India were acclimated to laboratory conditions in daily renewed fresh

water for fifteen days. Fish were fed ad libitum with a balanced diet prepared in the laboratory were maintained before the commencement of the experiment. Fishes were provided with adequate aeration. All healthy fishes alone selected and were maintained in the rectangular plastic tank of 70l capacity with conditioned parameters; temperature 28.5 ± 0.5º C, pH 7.8, dissolved oxygen 5.37 ppm. Fish weighing 70 ± 5 gm were used in all experiments. Fishes were provided with adequate aeration.

Preparation of aqueous methanolic extract

Selected plants are authenticated by Dr.Jothibasu., Assistant professor in Botany (DDE) Alagappa University, Karaikudi. Freshly collected leaves are washed with distilled water shade dried and ground as a coarse powder. Coarse powder was soaked with aqueous methanol in 1:3 ratio for 72hrs. It is filtered with pure muslin cloth and evaporated. After evaporation it was incorporated with the basal diet in various concentration.

Preparation of diets

The balanced diet prepared by adding fishmeal, soyabean meal, ricebran, wheat flour, vitamins and minerals in appropriate proportion. The three experimental diets namely E1, E2, E3, were prepared by adding the appropriate concentration of aqueous methanolic extracts of chosen medicinal plant. The diet E1-E3 were supplemented with aqueous methanolic extract of *Tridax procumbens*, of 0.01%, 0.1%, 1.0% respectively.

Collection of blood

Blood was collected from cardinal vein of the fish for every seven days interval in the EDTA coated blood collection tubes after the final feeding.

Total WBC Count

The total WBC was calculated using the formula (Larsen and Snieszko, 1964)

$$\text{Number of cells (cu.mm/1)} = \frac{\text{Number of cells counted x dilution}}{\text{Area counted x Depth of fluid}}$$

Phagocytic activity

The phagocytic activity assay was performed by the following modified method of Sahoo and Mukherjee (2002). Blood (100 µl) was mixed with equal quantity of bacterial suspension (1:1) in eppendorff tubes. The density of the bacterial culture was maintained throughout the experiment at 10^4 cells/ml in PBS. The mixture was incubated for 20 min at room temperature. After incubation, a thin smear was prepared and fixed with absolute alcohol for 5 min. The smear was later stained with Giemsa stain for 5 min and the phagocytic cells that have engulfed bacteria were counted (under microscope) as positive (Seeley et al., 1990).

$$\text{Phagocytic index (\%)} = \frac{\text{phagocytic leukocyte number}}{\text{observed total leukocyt number}} \text{ X } 100$$

Statistics mean, standard deviation, ANOVA tests

Tukey's Multicomparison test were performed in this study by using the SPSS software package. Differences were stastically significant at $p<0.05$ for the all the experiments used in this study.

Result

Total WBC count

The Total WBC count in fish administered orally with the different concentration of plants extract *Tridax procumbens* increased over that of the control. Thus the control value of 18.42 x 10^3/mm^3 increased to 22.16 x10^3/mm^3 , 23.18 x10^3/mm^3 and 24.20 x 10^3/mm^3 in the firstweek; 18.64x

$10^3/mm^3$ increased to 24.20 x$10^3/mm^3$, 25.26 x$10^3/mm^3$ and 28.20 x$10^3/mm^3$ in the second week; and 19.12 x $10^3/mm^3$ to 32.28 x $10^3/mm^3$, 32.66x$10^3/mm^3$ and 45.66x$10^3/mm^3$ in the third week; 19.08 x $10^3/mm^3$ to 32.30× $10^3/mm^3$, 36.66 ×$10^3/mm^3$ and 45.66×$10^3/mm^3$ in the fourth week after the fish administered orally 0.01%, 0.1%, 1.0% with the plant extract respectively. Thus, the Total WBC count increases in the fish were high in the third week than the increases in the first, second and fourth week of the experiment. Thus, this study highlights that the 1% of plants extract has enhanced the Total WBC count to a highest level in the third week of the experiment (Fig.1). Total WBC count was significantly ($p<0.05$) higher in all the fish administered orally with plant extracts at all the assay period as compared to the control group. Moreover, the significant increase of WBC count ($p<0.05$) was observed maximum in the fish in the third week of the experiment. Interestingly more or less the same total WBC count was maintained in the fourth week. This result clearly indicates that the Total WBC count was depending on concentration of extract and duration of the experiment.

Phagocytic activity

The phagocytic activity in fish administered orally with different concentrations of aqueous methanolic extract of *Tridax procubens* increased over that of control. Thus the control value of 26.00 % increased to 27.44%, 28.00 % and 29.22 % in the first week; 26.00% increased to 29.60%, 29.34% and 32.46% in the second week; and 27.00% increased to 32.00%, 39.00 % and 55.44% in the third week; 27.44% to 32.00%, 39.22%, 55.44% in the fourth week after the fish administered orally with 0.01%, 0.1%, 1.0% of plants extract respectively. Thus, the phagocytic activity increase was the maximum in the third week than the increase in the first, second and fourth week of experiment. Highest 1% of the plant extract showed a best result than other doses of aqueous methanolic extract of *Tridax procumbens*. However, this increase is the maximum in the third week of oral administration of fish with plants extract in *C. sriatus*. The phagocytic activity significantly ($p<0.05$) increased in all experimental groups compared with the control group. The significant ($p<0.05$) elevation of phagocytic activity was recorded in the fish fed orally with 1% of plant extract in third week of the experiment. This result clearly indicates that the phagocytic activity was depending on concentration of extract and duration of the experiment.

Discussion

Total WBC count

The percentage of increase in the total WBC count of fish administered orally with the plant extract is 69% when compared with the control group. This corroborates with the total WBC count increase in fish by the different plant extracts; 59%increase in *Cyprinus carpio* with *Euphorbia hirta* plant leaf extract **(**Pratheepa and Sukumaran, 2014). 76% increase with *Solanum trilobatum* and *Ocimum sanctum* in *Mystus keletius* (Subeenabegum and Navaraj, 2016); 65% in *Cyprinus carpio* fed with dietary levamisole (Maqsood, 2009); 56% in carp immunized with Ganaderma and Astragalus (Yin *et al.*, (2009); in rainbow trout, *Oncorhynchus mykiss* after 2 weeks feeding with garlic and *Allium cepa* (Nya and Austin, 2009); in *O. mossambicus* after 3 weeks feeding with *Tinospora cordifolia* leaf extracts (Alexander, 2007); in Labeo rohita after 3 weeks feeding with 0.5% (or) 1% of *Allium sativum* (Sahu et

al., 2006); Thus the Total WBC count increase in the fish may be due to an initial sign of non-specific immune response (Manjrekar *et al.*, 2000; Christybapita *et al*., 2007) or may be due to interdependent mechanism of an innate resistance and adaptive immunity (Mishra *et al*., 2009) that may enhance the Total WBC count, as it is the first line of defense (Mydeen and Haniffa, 2011). The highest level of phagocytes observed only in the third week. It is due to the digestive adaptability. This finding has gained the support from an enhancement in monocytes, granulocytes, neutrophils and macrophages in fish fed with *Solanum trilobatum* (Bouic *et al*., 1996)

Phagocytic activity

The percentage of increase in the phagocytic activity of fish administered orally with the plant extract is 71% in *Tridax procumbens* when compared with the control group. The oral administration 1% of the plant extract elevates the phagocytic activity in the fish in the third week of the experiment. This finding corroborates with the findings of; 64% *Cyprinus carpio* with *Euphorbia hirta* plant leaf extract **(**Pratheepa and Sukumaran, 2014)., in hybrid tilapia fed with a diet supplemented with garlic for 2 weeks (Ndong and Fall, 2011); 1% of *Allium sativum* (Sahu *et al*., 2006); in tilapia fed with a diet supplemented with *Allium sativum* for 4 weeks (Govind *et al*., 2012); in rainbow trout fed with 1% aqueous extract of powdered ginger root for 3 weeks (Dugenci *et al*., 2003); in parrot fish *Oplegnathus fasciatus* fed with diet formulated with DL α tocopheryl acetate diet for 12 weeks (Galaz *et al*., 2010); 70.33% increase *O. sanctum* in fingerlings of *Cyprinus carpio* with extract of *Andrographis paniculata* and *Acalypha indica* (Muthumurugan *et al*., 2007); 58% in *Solanum trilobatum* and *Ocimum sanctum* in *Mystus keletius* (Subeenabegum and Navaraj, 2016); 65.0% increase in *C. carpio* with leaf extract of *Ocimum sanctum* (Pavaraj et al., 2011); 68.4% increase in *C. carpio* with the leaf extract of *S. trilobatum* (Durgadevi and Balasubramanian, 2009). Moreover, the enhancement of the phagocytic activity in fish may be due to the first level defense against pathogen (Manjrekar *et al.*, 2000) or may be due to the proliferation of lymphocytes which is under the influence of gene expression of cytokine (Wang et al., 1997) or may be due to the stimulation by macrophages (Tang *et al*., 2010) or due to an increase in lysozyme activity and other humoral factors (Chung and Secombes, 1987) or due to an increase in neutrophilic granulocytes (Secombes and Fletcher1992) ; and Vankemenade *et al*., 1995 Thus, the maximum phagocytic activity in the fish was elevated by the plant extract *Tridax procumbens* when compare with control group. The highest level of phagocytes observed only in the third week due to the digestive adaptability. Hence, this trend may be due to the immunostimulatory effect of the phytocomponents in the aqueous methanolic extract of *Tridax procumbens*.

Conclusion

Aqueous methanolic extract of Tridax procumbens in this study considerably has enhanced the non- specific immune response in *C. striatus*. Anyhow in addition, the underlying molecular mechanism beside the isolation and characterization of the active compounds from the plants require more studies.

Acknowledgement

Our very sincere and heartiest thanks to the Management of Alagappa University, Karaikudi for their strong support to carry out this valuable research. Our sincere thanks to the Dr.B.Vaseeharan, Professor and Head, Department of Animal Health and Management, Alagappa University, Karaikudi.

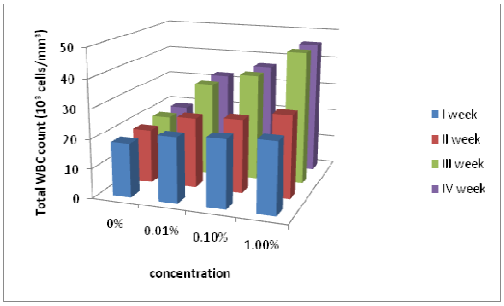

Fig. 1: Graphical presentation showing Total WBC count in oral administration of *Tridax procumbens* in *C.striatus* from Ist to IVth week (Mean ± S.D)

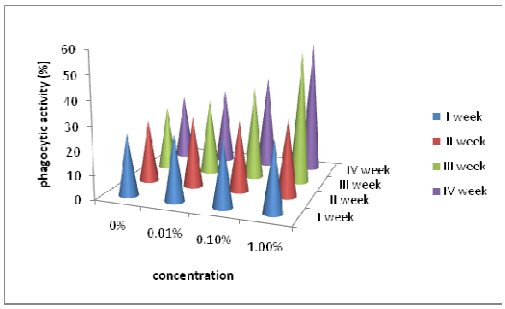

Fig. 2: Graphical presentation showing Phagocytic activity in oral administration of *Tridax procumbens* in *C. striatus* from Ist to IVth week (Mean ± S.D)

References

Alexander, CP. Studies on immunomodulation by *Tinospora cordifolia* (Miers) leaves in *Oreochromis mossambicus* (Peters) 2007. Ph.D: Thesis Madurai Kamaraj University Madurai, Tamil Nadu.

Anderson, DP. Immunostimulants adjuvants and vaccine carriers in fish application to aquaculture. Annual Review of Fish Diseases 1992; 2:281–307.

Balasubramanian, V, Physiology of *Cyprinus carpio* infected by *Aromonas hydrophila* and efficacy of a selected neem formulation to cure the disease. 2006; Ph.D, Thesis: M.K. University, Madurai, Tamil Nadu, India.

Barlas, A. Post–harvest technology for fish. Progressive Farm 1986; 6:59 – 65.

Barker, D, Holliday, J. Blood lymphocyte and neutrophil response of cultured rainbow trout *Oncorhynchus mykiss* administered varying dosages of an oral immunmodulator-Fin-immune TM. International Journal of Biological and life sciences2009;1: 9-13.

Bouic, PJD, Etsebeth S.Beta-sitosterol glucoside stimulate human peripheral blood lymphocyte Proliferation, Implication for their Use as an Immunomodulatory Vitamin Combination. Int. J. Immunopharmacology 1996; 12:693-700.

Chansue, N, Ponpornpisit A, Endo M, Sakai M, Satoshi . Improved immunity of tilapia *Oreochromis niloticus* by C-UP III, a herb medicine. Fish Pathol 2000; 35: 89-90.

Citarasu, T, Venkatramalingam K, Babu MM, SekarRJ R, Petermarian M. Influence of the antibacterial herbs *Solanum trilobatum*, *Andrographis paniculata* and *Psoralea corylifolia* on the survival growth and bacterial load of *Penaeus monodon* post larvae Journal Article. 2003; 11:581-595.

Divyagnaneswari, M, Christybapita R, Michael RD. Enhancement of nonspecific immunity and disease resistance in *Oreochromis mossambicus* by *Solanum trilobatum* leaf fractions Fish Shell fish Immunol 2007;23:249–259.

Dugenci, SK, Arda N, Candan A. Some medicinal plants as immunostimulant for fish J. Ethnopharmacol 2003; 88: 99-106.

Durgadevi, G, Balasubramanian V. Studies on the immunoprotective effect of *Solanum trilobatum* on the common carp *Cyprinus carpio (L.).* Proceeding of the conference on Recent Advances in Applied Zoology UGC and the management of ANJAC 2009; 90-93.

Engstadt, R, Robertson, B. Recognition of yeast wall glucan by Atlantic salmon (*Salmo salar*). Dev Comp Immunology 1993; 17:319-330.

Goetz, FW, Iliev, DB, McCauley, LAR, Liarte, CQ, Tort LB, Planas, JV. Analysis of genes isolated from lipopolysaccharide-stimulated rainbow trout (*Oncorhynchus mykiss*) macrophages. Molecular immunology 2004; 41: 1199-1210.

Harikrishnan, R, Rani, CN, Balasundaram, C. Hematological and biochemical parameters in common carp *Cyprinus carpio* following herbal treatment for *Aeromonas hydrophila* infection Aquaculture. 2003; 221: 41–50.

Jeney, G, Anderson, DP. An invitro technique for surveying immunostimulants in fish. Aquaculture 1993; 112: 283–287

Kaleeswaran, B, Ilavenil S, Ravikumar S. Screening of phytochemical properties and antibacterial activity of *Cynodon dactylon (L.).* Int J Curr Res 2010; 3:83-88.

Larsen, H, N, Snieszko SF. Modification of the microhematocrit technique with trout blood. Trans American Fish Soc 1964; 90: 139 -42.

Lipton. Marine secondary metabolites (MSM) enhance bacterial clearance in *Penaeus monodon.* The Israeli Journal of Aquaculture 2009; 6:42-47.

MacArthur, JI, Thomson, AW, Fletcher, TC. Aspects of leucocyte migration in the plaice *Pleuronectes platessa L.*. J Fish Biol 1985; 27:667–676.

Magnadottir, B. Innate immunity of fish (Overview). Fish Shellfish Immunol 2006; 20: 137-151.

Marzouk, MS, Moustafa, MM, Mohammed, NM. Evaluation of immunomodulatary effects of some probiotics on cultured *Oreochromis niloticus.* Aquaculture 2008; 43:1043-1058.

Masoud Haghighi and Mostafa Sharif Rohani. The effects of powdered ginger (*Zingiber officinale*) on the haematological and immunological parameters of rainbow trout *Oncorhynchus mykiss*. Journal of medicinal plant and herbal therapy research. 1 (2013): 8-12.

Mohamad, S, Abasali, H. Effect of Plant Extracts Supplemented Diets on Immunity and Resistance to *Aeromonas hydrophila* in Common Carp (Cyprinus carpio). J Agri 2010; 5:119-127.

Nya, EJ, Austin B. Development of immunity in rainbow trout (*Oncorhynchus mykiss*) of *A. salmonicida* antigens. Veterinary Immunology and Immunopathology 2009;19:153–164.

Olivier, G, Eaton, CA, Campbell, N. Interaction between *Aeromonas salmonicida* and peritoneal macrophages of brook trout *Sallerinus fontinalis*. Vet Immunol Immunopathol 1986;12:223–234.

Pavaraj, M, Balasubramanian, V, Baskaran S, Ramasamy P. Development of Immunity by Extract of Medicinal Plant *Ocimum sanctum* on Common Carp *Cyprinus carpio* (L.). Research Journal of Immunology 2011; 4:12-18.

Pratheepa, V**,** Sukumaran, N. Effect of Euphorbia hirta plant leaf extract on immunostimulant response of *Aeromonas hydrophila* infected *Cyprinus carpio.* 2014. 13; 2; 671-674.

Raa, J, Marie D, Tapia-Salazar M, Novoa OMA, Cerecedo CR. The use of immunostimulants in fish and shellfish feeds. In: Cruz-Suarez LE, Ricque,editors. Advances en nutricion acuicola V, Memorias V, del V. Simposium International de Nutricion Acuicolan Merida Yucatan Mexico 2000; 19-22.

Raa, J, Rorstad, G, Engstad, RE, Robertson B. The use of immunostimulants to increase resistance of aquatic organisms to microbial infections. Disease in Asian aquaculture. Proceedings of the first symposium on diseases in Asian aquaculture. Philippines: Asian Fisheries Society 1992; 39-50

Rajesh, KC, Subhah, KP, Prasad, AK, Pal, K. Evaluation of *Bacillus subtilis* as a probiotic to Indian major carp Labeo rohita. Aquaculture 2006; 37: 1215-1221.

Rao YV, Das BK, Jyotyrmayee, P, Chakrabarti R. Effect of *Achyranthes aspera* on the immunity and survival of Labeo rohita infected with *Aeromonas hydrophila.* Fish Shellfish Immunol 2006; 20:263–273.

Ravenhalt, A. Malnutrition in Philippines UFSI. Rep South East Asia Ser 1982; 20:73-84.

Sahoo, PK, Mukherjee, SC. 2002. Influence of high dietary α-tocopherol intakes on specific immune responses, non specific resistance factors and disease resistance of healthy and aflatoxin B1-induced immunocomprmised Indian major carp. *Labeorohita* (Hamilton) Aquaculture Nutrition 8:159-167.

Sahoo, PK, Mukherjee, SC. Immunomodulation by dietary vitamin C in healthy and aflatoxin β-1 induced immunocompromised rohu (*Labeo rohita*). Comp Immunol Microbiol Infect Dis 2003; 17:26-76.

Sakai, M. Current research status of fish immunostimulants. Aquaculture 1999; 172:63–92.

Selvaraj, V, Sampath, K, Sekar, V. Adjuvant and immunostimulatory effects of ßglucan administration in combination with lipopolysaccharide enhances survival and some immune parameters in carp challenged with *Aeromonas hydrophila*. Vet Immunol Immunopathol 2006; 114:15-24.

Sharma, S, Ashutosh., A, Riteshkumar., T, Chanu., TB, Das., A. Effect of *Withania somnifera* (L.Dunal) root as a feed additive on immunological parameters and disease resistance to *Aeromonas hydrophila* in Labeo (*Hamilton*) fingerlings. Fish Shell fish immunol 2010; 29:508-512.

Subeena begum, S and Navaraj PS. Studies on the immunostimulatory effect of extract of *Solanum trilobatum* and *Ocimum sanctum* in *Mystus keletius*. Inter national journal of fisheries and aquatic studies 2016; 4(2): 376-381.

Sudhakaran SD, Srirekha P, Devasree S, Premsingh S, Michael RD. Immunostmulatory effect of *Tinospora corfolia* miers leaf extracts in *Oreochromis mossambicus*. Indian J Exp Biol 2006;44:726-732.

Venkatachalam Uthayakumar, Immunostimulatory effect and disease resistance induced by *Lawsonia inermis* against *Aphanomyces invadans* in striped murrels (*Channa striatus*). 2014 ; 1(4):231-241.

Wafaa A, Hasan MA, Nefisa A, Shayeb E. Biological and anti-microbial activities of aqueous extracts from neem tree (*Azadirachta indica*). J App Sci Res 2007; 3:1050-55.

Wu CC, Liu CH, Chang YP, Hsieh SL. Effects of hot-water extract of *Toona sinensis* on immune response and resistance to *Aeromonas hydrophila* in *Oreochromis mossambicus*. J Fish Shellfish immunol 2010; 29:258-263.